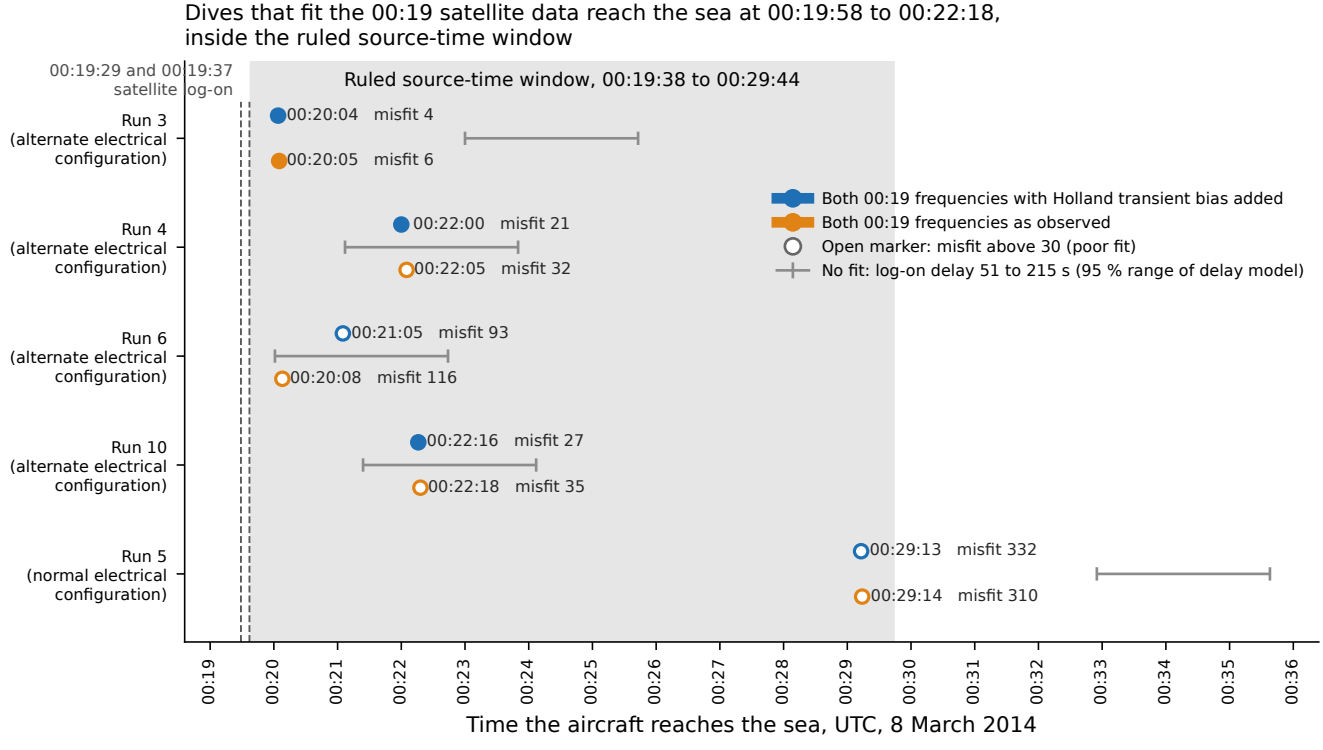

Notes. Source: Boeing 2016 engineering-simulator runs (traces released 19 Aug 2018), 1 s position and altitude, no pilot input after fuel exhaustion. Main electrical power is lost at the first engine stop (alternate configuration) or the second engine stop (normal configuration, run 5). Each run is placed with its 00:10:59 point on the 00:10:59 timing arc (41 to 19.5 S, 0.1 degree steps), its track turned -30 to +30 degrees, and its power loss set 30 to 510 s before 00:19:29.4 (1 s steps; the satellite unit answered at 00:10:59, so power was on then). Misfit = chi-squared on the burst timing offsets at 00:19:29 (sigma 63 us) and 00:19:37 (43 us) and the burst frequency offsets at 00:10:59, 00:19:29 and 00:19:37 (sigma 7 Hz), with the frequency bias fitted under a 150 +/- 5 Hz prior. Holland transient bias: fitted 17 to 136 Hz at 00:19:29, 0 to 6 Hz less at 00:19:37 (Holland 2017, arXiv:1702.02432). Bars: all delays within a misfit of 4 of the best (log-on delay model not applied); dot: best fit. Alignments with the first engine stop before 00:10:59 are kept only if the descent rate then is 800 ft/min or less. Sea contact = end of record (473 to 1,131 ft) plus the remaining height at the final 10 s descent rate. Simulator wind is kept. Late-dive speeds near Mach 1.0 are outside Boeing's simulator data. Run 5 has no fit (misfit 296 to 332, latitude at the grid limit) and is a sensitivity only. Grey bars: impact time for a log-on delay from the gamma(8, 14.875 s) delay model, without the satellite data. Normal runs 1, 2, 7, 8 and 9 reach the sea at 00:33:26 or later for every allowed delay (off this chart).